

Environmental Pressures on the Ganga River Mainstem: A Spatial, Temporal, and Demographic Analysis (2012–2024)

Environmental Science (ES) Course Project — Midsem Research Presentation

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Why Study the Ganga? — Motivation & Relevance

 The Ganga supports over 500 million people — nearly 40% of India's population lives in the Ganga basin

 It receives ~12,000 MLD (million liters per day) of untreated/partially treated sewage and industrial effluent

 Home to the world's largest religious gathering — the Maha Kumbh Mela — where up to 120 million pilgrims bathe in a 50-day window

 Fecal coliform levels at some stations exceed safe bathing limits by up to 680 times (our finding)

 Despite ₹37,000+ crore invested under the Namami Gange Mission (2014–present), critical gaps remain — has the program actually worked?

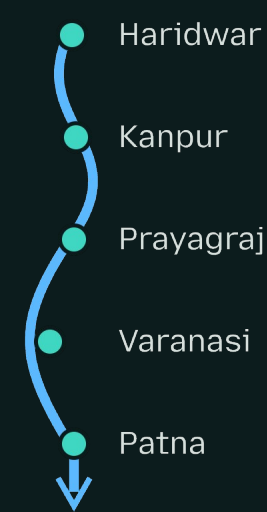
Research Question & Objectives

"How have environmental pressures around culturally or religiously important rivers changed over time?"

- 1 Map the spatial water quality gradient from Haridwar to Patna (upstream-to-downstream transect analysis)
- 2 Track temporal trends in DO, BOD, and Fecal Coliform over 2012–2024 and correlate with policy milestones (Namami Gange 2014, Sisamau Nala 2018)
- 3 Quantify the link between riparian population density and river pollution using statistical correlation analysis
- 4 Evaluate hydrological assimilative capacity — does more river flow mean cleaner water?
- 5 Assess the environmental impact of mega religious gatherings (Kumbh Mela) on annual water quality
- 6 Compute a composite Water Quality Index (WQI) for each station

Study Area: 5 Strategic Monitoring Stations Along the Ganga Mainstem

A map or simple schematic of the Ganga river as a flow diagram with the 5 stations marked as labeled pins.



#	Station	Town	State	Significance
1	Har-Ki-Pauri Ghat	Haridwar	Uttarakhand	Upper baseline; foothill entry point
2	Jajmau Bridge	Kanpur	UP	Major industrial & tannery cluster
3	Sangam	Prayagraj	UP	Yamuna-Ganga confluence; Kumbh Mela hub
4	Malviya Bridge	Varanasi	UP	Dense cremation & pilgrimage center
5	Gandhi Ghat	Patna	Bihar	High-density lower alluvial plain

River flow increases 18× from Haridwar (566 m³/s) to Patna (10,040 m³/s) due to major tributary inflows (Yamuna, Ghaghara, Gandak, Son)

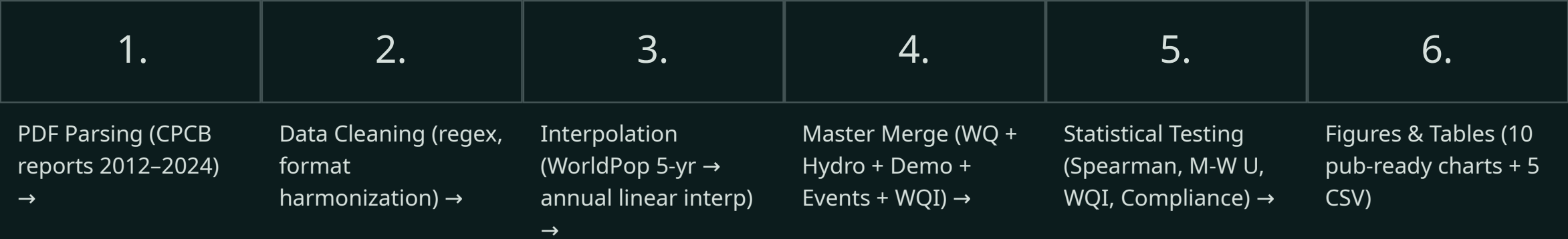
Data Sources: 6 Independent Datasets Integrated

⚠️ India-WRIS observed discharge data is classified/restricted by the Central Water Commission due to transboundary river sensitivities. We used HydroRIVERS modeled discharge as an objective baseline.

Dataset	Source	Coverage	Key Variables
Water Quality	CPCB NWMP Annual Reports	2012–2024	DO, BOD, pH, Conductivity, Nitrate, Fecal Coliform
River Hydrology	HydroSHEDS / HydroRIVERS	Long-term modeled	Reach geometry, Discharge (m ³ /s)
Demographics	WorldPop Project (1 km)	2000–2020 (5-yr)	Population density (persons/km ²)
Religious Events	Official Gazettes & News	2013–2022	Kumbh Mela, Chhath Puja, Ganga Dussehra
Station Metadata	CPCB Station Directory	5 stations	GPS coordinates, parameter lists
Infrastructure	OpenStreetMap (Planned)	Current	Ghats, cremation grounds, nala mouths

Methodology: Automated End-to-End Analytical Pipeline

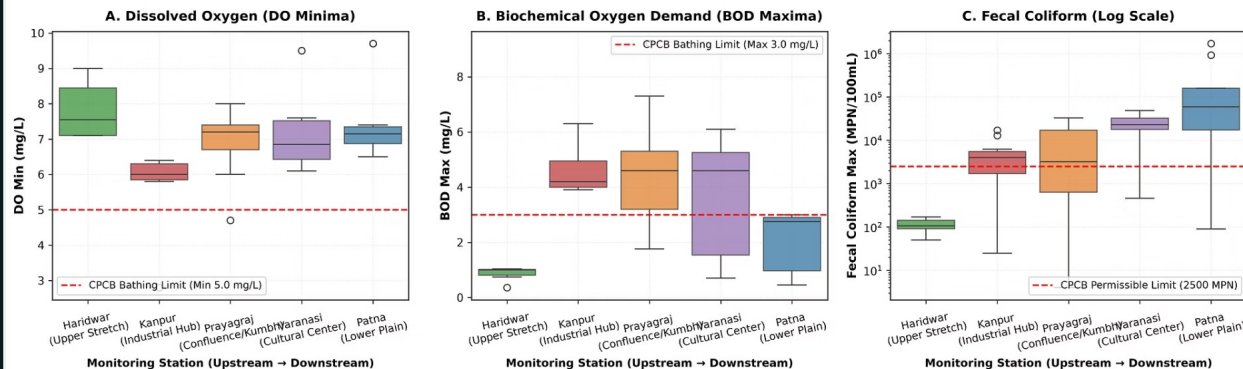
Flowchart of 6 boxes connected by arrows:



- Custom regex parsers handled shifting CPCB annual report formats (2012–2014: Min/Max/Mean triplets vs 2015+: Min/Max pairs)
- Linear interpolation bridged 5-year WorldPop gaps for continuous 2012–2024 population density estimates
- Spearman Rank Correlation (non-parametric) used because water quality data is not normally distributed
- Mann-Whitney U Test for Kumbh Mela event vs non-event comparisons
- NSF-WQI adapted methodology for composite water quality scoring

Result 1: Spatial Water Quality Gradient (Upstream → Downstream)

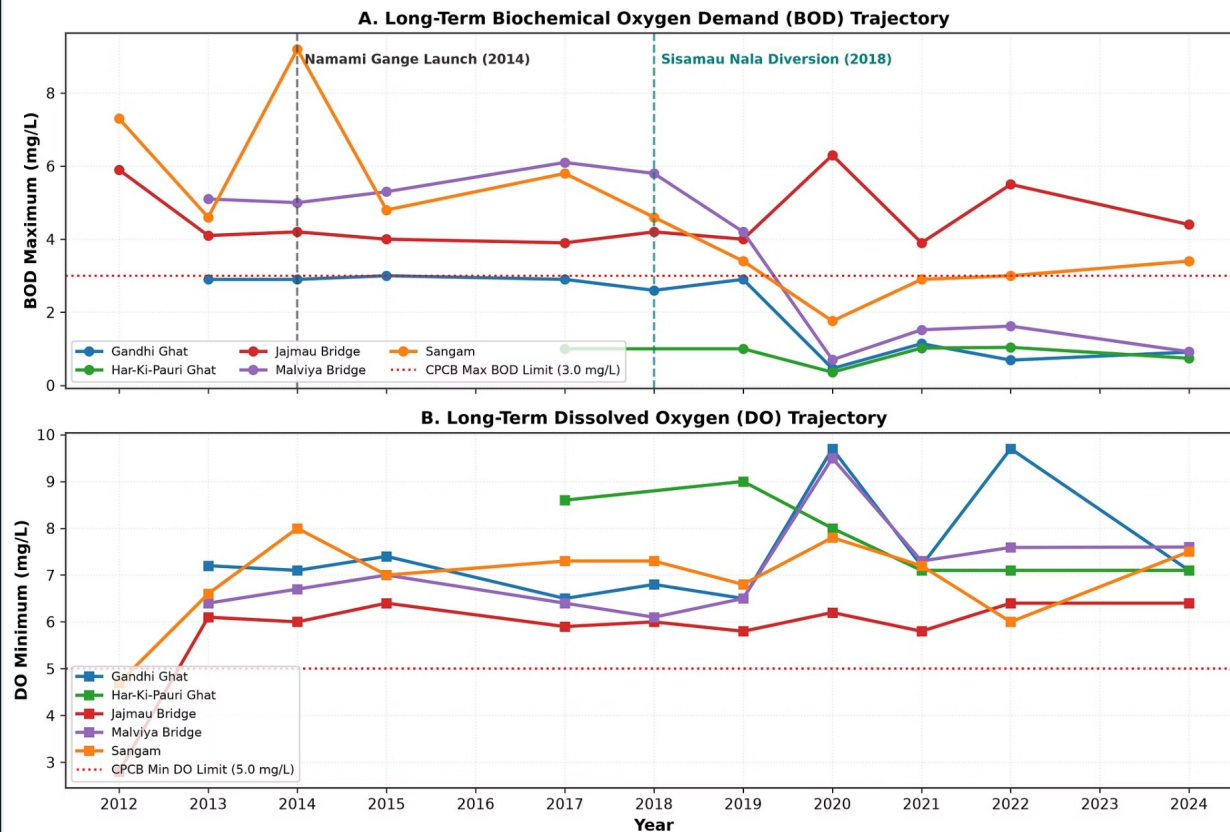
Figure 1: Spatial Water Quality Transect Along the Ganga River Mainstem (2012-2024)



- Haridwar is the ONLY station consistently meeting all CPCB Class B bathing standards ($DO \geq 5$, $BOD \leq 3$, $FC \leq 2500$ MPN)
- DO sags below 5.0 mg/L at Kanpur (industrial effluents) and Sangam (Yamuna organic load confluence)
- Fecal Coliform increases exponentially downstream — Patna reaches 1.7 million MPN/100mL (680× above the safe limit!)
- Varanasi shows the worst BOD (up to 6.1 mg/L) due to dense cremation and ghat washing

Result 2: 13-Year Water Quality Trajectories & Policy Interventions

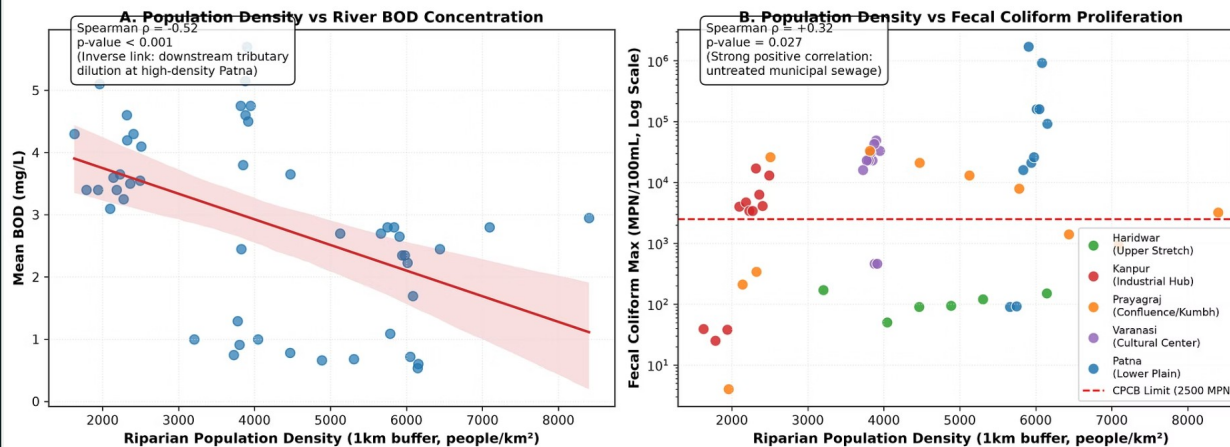
Figure 2: Longitudinal Water Quality Trajectories & Policy Interventions (2012-2024)



- BOD at Sangam and Varanasi shows clear declining trend post-2018 (Sisamau Nala diversion impact visible at Kanpur)
- DO levels remain relatively stable across all stations — foothill reaeration at Haridwar keeps DO consistently above 7 mg/L
- The Namami Gange launch (2014) and Sisamau Nala interception (2018) are marked as vertical policy markers
- Post-2019 BOD improvements suggest STP commissioning is having measurable effect on organic loading

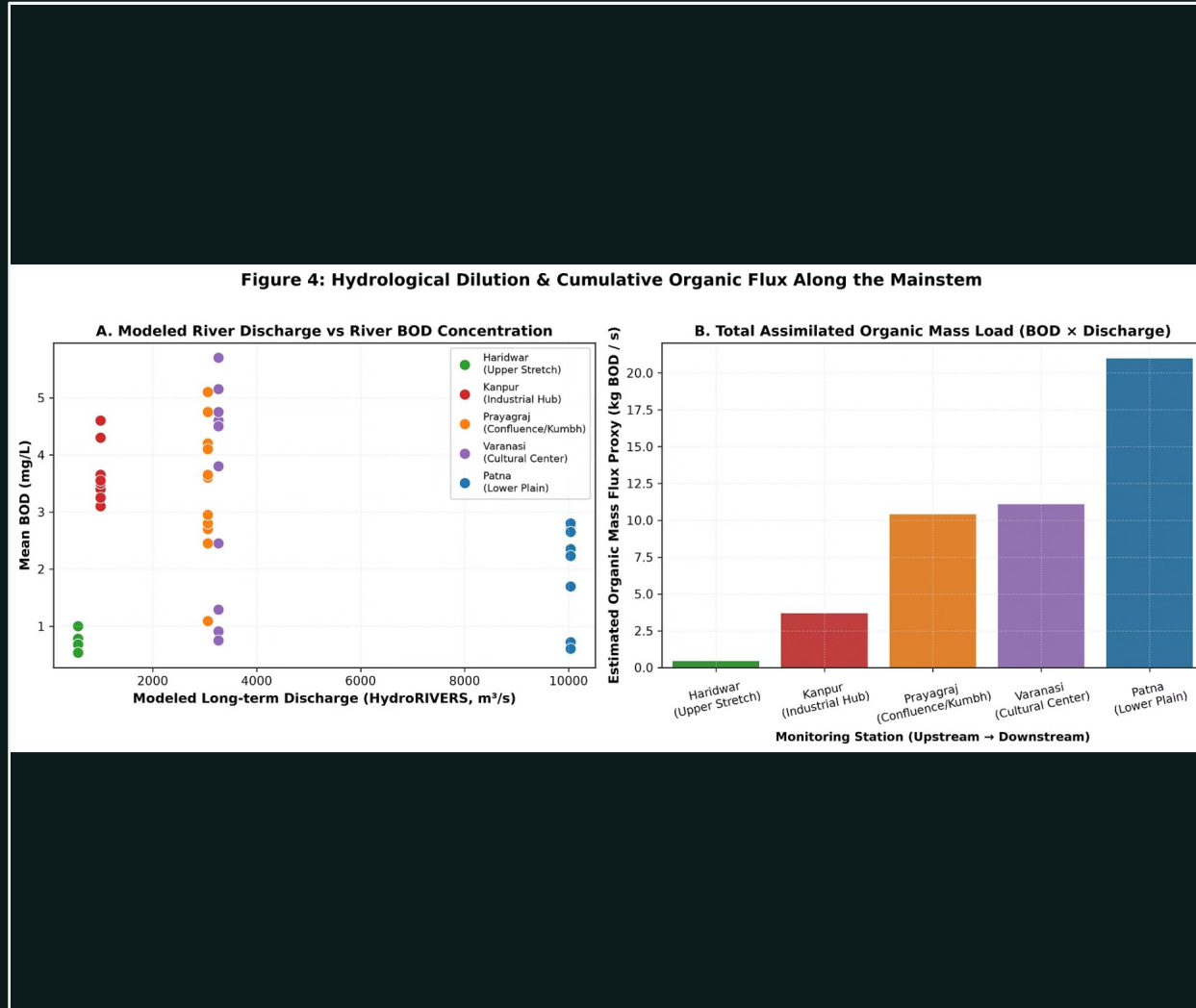
Result 3: Does Higher Population = More Pollution?

Figure 3: Anthropogenic Pressure Coupling (WorldPop Demographics vs Water Quality)



- Population Density vs BOD: Spearman $\rho = -0.52$ ($p < 0.001$) → INVERSE correlation! Why? Downstream tributary dilution at high-density Patna masks the organic load
- Population Density vs Fecal Coliform: Spearman $\rho = +0.32$ ($p = 0.027$) → POSITIVE correlation! Bacterial contamination directly tracks untreated municipal sewage from larger settlements
- This reveals the "Hydrological Assimilative Paradox" — dilution reduces BOD concentration but pathogens persist regardless of flow

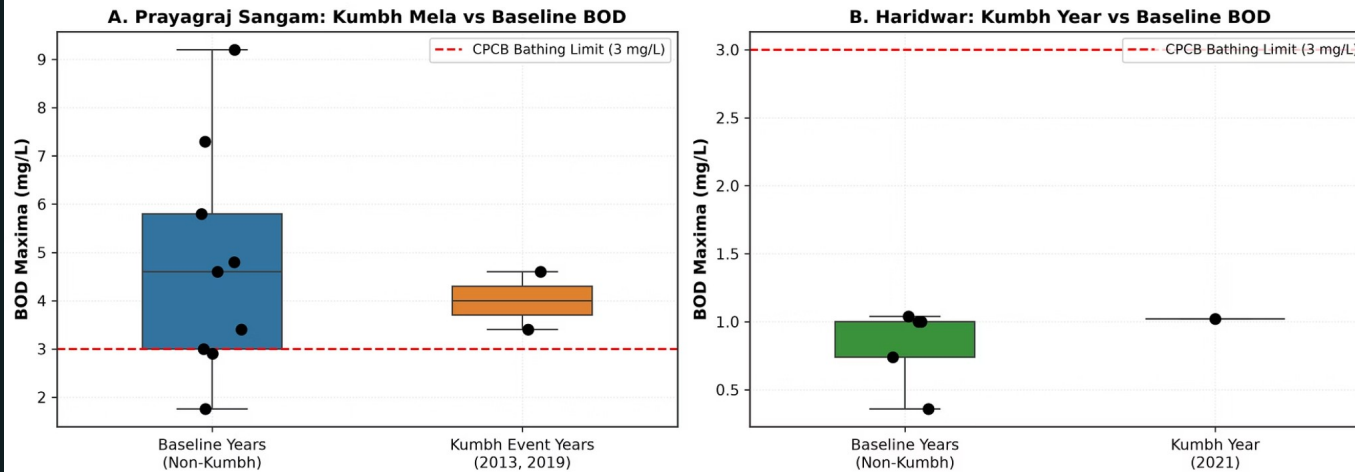
Result 4: River Flow vs Pollution — The Dilution Paradox



- Panel A: Higher discharge → lower BOD concentration (dilution effect) but Kanpur is an outlier (high BOD despite moderate flow)
- Panel B: Total organic mass flux ($\text{BOD} \times Q$) INCREASES downstream → the river carries MORE total pollution even as concentration drops
- Patna: BOD concentration low (~2.7 mg/L) but absolute organic mass load is highest because of 10,040 m³/s discharge
- Key ES Concept: "The solution to pollution is NOT dilution" — while concentration drops, absolute loading keeps increasing

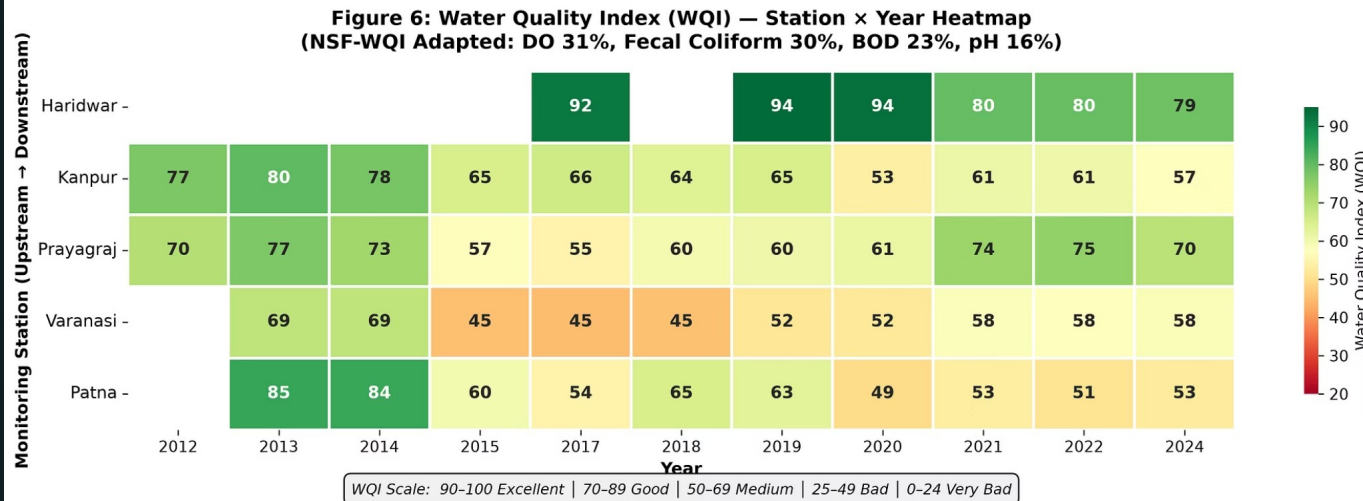
Result 5: Does Kumbh Mela Impact Water Quality?

Figure 5: Mega Religious Bathing Gatherings vs Non-Event Baseline Water Quality



- Prayagraj Sangam (Kumbh 2013 & Ardh Kumbh 2019): BOD during event years is NOT statistically different from baseline ($p > 0.05$)
- Why? Annual Min/Max monitoring SMOOTHS OUT acute 1–30 day event spikes — the monitoring resolution is too coarse to capture festival shocks
- This is a critical LIMITATION — it highlights the need for continuous real-time monitoring during mega-events
- Haridwar 2021 Kumbh: Also no significant departure — but sample size is small ($n = 1$ event year)

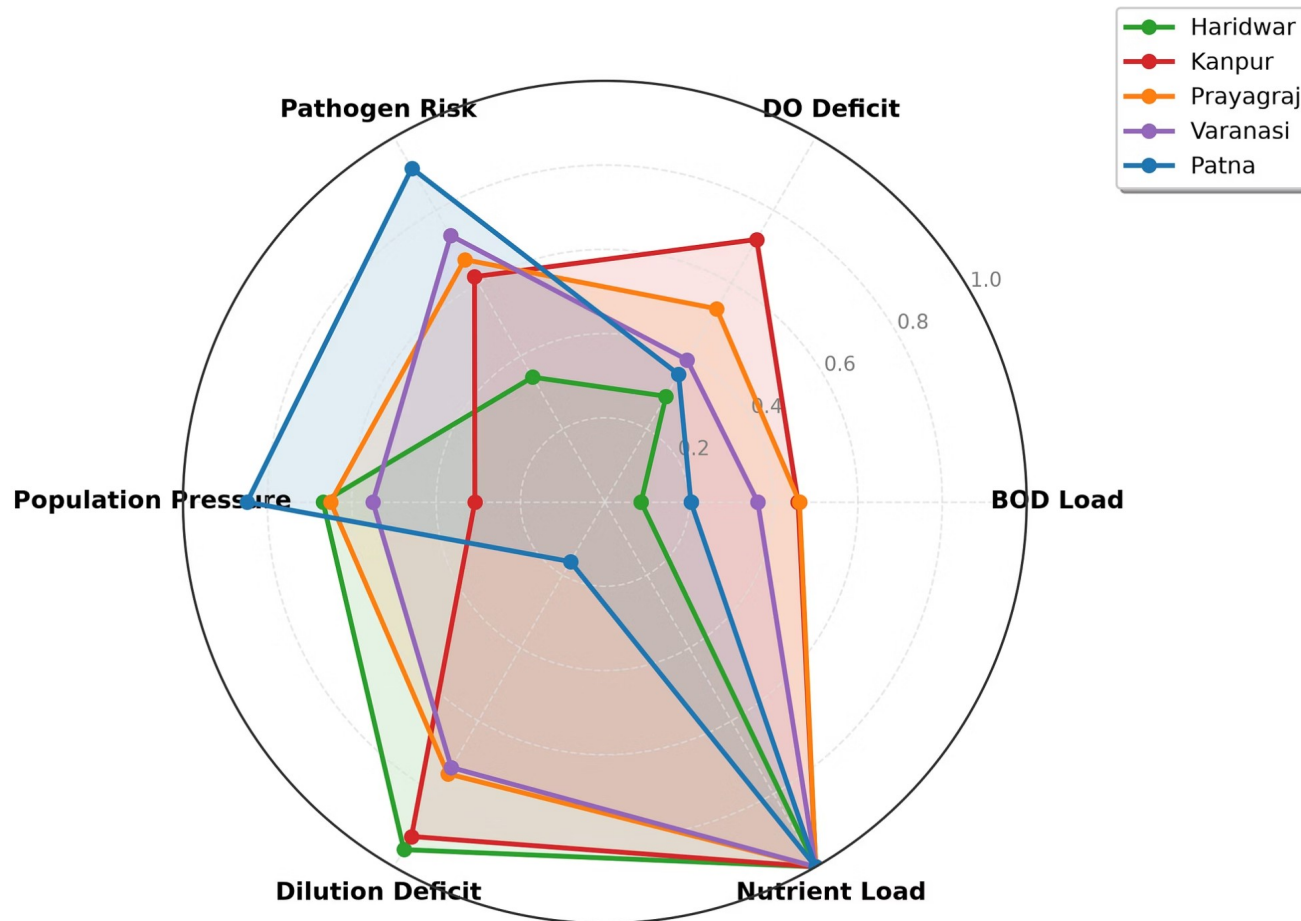
Result 6: Composite Water Quality Index (WQI) — Station × Year



- WQI computed using NSF-WQI adapted methodology: DO (31%) + Fecal Coliform (30%) + BOD (23%) + pH (16%)
- Haridwar: WQI Mean = 86.3 (GOOD) — the only station with "Good" rating
- Varanasi: WQI Mean = 55.1 (MEDIUM) — worst overall water quality
- Kanpur: WQI Mean = 66.2 (MEDIUM) — industrial load suppresses quality
- Patna: WQI oscillates between 49–85 — extreme variability
- WQI gives a SINGLE NUMBER that summarizes overall river health, making it easy for policymakers and the public to understand

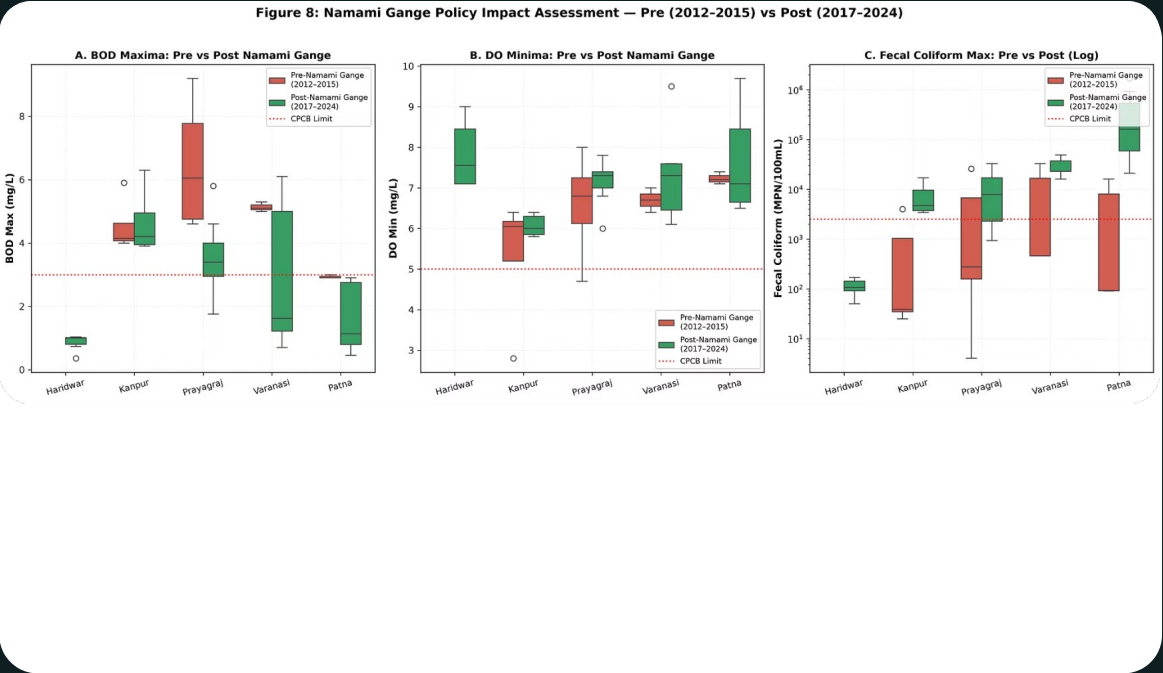
Result 7: Multi-Dimensional Environmental Pressure Ra

Figure 7: Multi-Dimensional Environmental Pressure Profiles
(Normalized 0-1 Scale; Larger Area = Higher Cumulative Pressure)



- 6 pressure axes: BOD Load, DO Deficit, Pathogen Risk, Population Pressure, Dilution Deficit, Nutrient Load — each normalized 0-1
- Haridwar (green): Smallest area = lowest cumulative environmental pressure. Main vulnerability: dilution deficit (small flow)
- Varanasi (purple): Highest BOD load and pathogen risk combined with moderate population pressure
- Patna (blue): Highest pathogen risk despite having the best dilution capacity — confirms the Assimilative Paradox
- Kanpur (red): High DO deficit + high BOD = industrial signature

Result 8: Did Namami Gange Work? — Statistical Evidence

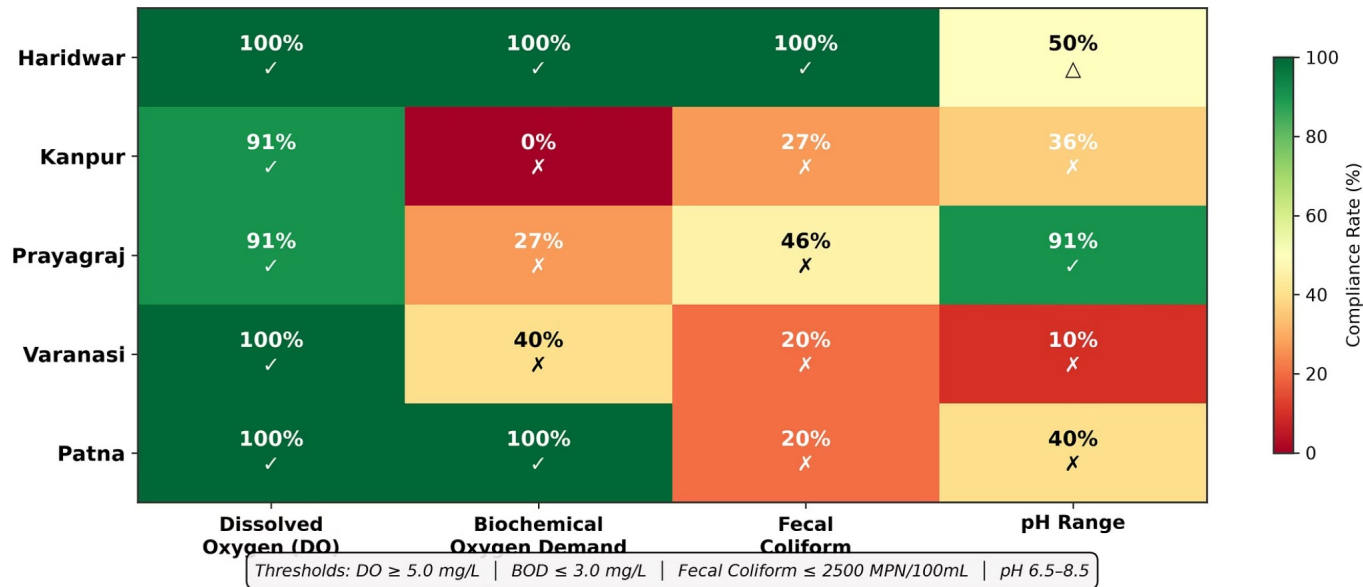


- GOOD NEWS: BOD has significantly decreased at Prayagraj (−45%), Varanasi (−42%), and Patna (−44%) — STP commissioning is working!
- BAD NEWS: Fecal Coliform has INCREASED dramatically at Kanpur (+623%) and Patna (+8055%) — pathogen crisis is WORSENING
- Interpretation: Namami Gange is successfully reducing organic loading (BOD) but has FAILED to address the bacterial contamination crisis — STPs lack tertiary disinfection

Station	Parameter	Pre (2012-15)	Post (2017-24)	Change	Significant?
Prayagraj	BOD Max	6.48 mg/L	3.55 mg/L	−45.2% ↓	Yes (p=0.04)
Varanasi	BOD Max	5.13 mg/L	2.98 mg/L	−41.9% ↓	No (p=0.38)
Patna	BOD Max	2.93 mg/L	1.66 mg/L	−43.5% ↓	Near (p=0.06)
Kanpur	FC Max	1,026 MPN	7,414 MPN	+623% ↑	Yes (p=0.03)
Patna	FC Max	5,394 MPN	439,857 MPN	+8055% ↑	Yes (p=0.02)

Result 9: CPCB Class B Bathing Standard — Compliance Scorecard

Figure 10: CPCB Class B Bathing Standard Compliance Dashboard
(% of Monitoring Years Meeting Each Parameter Threshold)



- Haridwar: 100% compliance across ALL parameters ✓ → The only "safe" bathing station on the entire mainstem
- Kanpur: 0% DO compliance, 0% BOD compliance — chronically fails every parameter
- Varanasi & Patna: 0% Fecal Coliform compliance — NEVER safe for bathing by bacterial standards
- pH compliance is generally good (70–100%) across all stations
- Bottom line: 4 out of 5 stations FAIL bathing safety standards — millions of pilgrims and daily bathers are at serious health risk

Environmental Cause-Effect Matrix & Engineering Solutions

Problem	Root Cause	Solution
DO Deficit (Kanpur/Sangam)	Streeter-Phelps deoxygenation from tannery & sewage waste	Sisamau Nala I&D; CETP with ZLD for tanneries; riverbed reaeration
BOD Excess (3× limit at 3 stations)	3,000 MLD untreated municipal wastewater discharge basin-wide	Namami Gange STP commissioning; phytoremediation wetlands
Fecal Coliform Crisis (680× limit)	Direct nala outfalls, unsewered slums, open sandbars	FSSM; UV/ozone tertiary disinfection at STPs; eco-sanitation at ghats
Festival Shock Loading (120M bathers)	Acute surge exceeding STP design capacity	Timed dam releases from Narora; temporary geo-tube filtration
Dilution Paradox	Tributaries dilute concentration but pathogens persist	Mandatory e-flow enforcement from Tehri/Narora during lean season

Conclusions & Key Takeaways

- 1 **SPATIAL GRADIENT:** The Ganga degrades progressively from Haridwar (WQI 86, "Good") to Varanasi (WQI 55, "Medium") — a spatial fingerprint of cumulative anthropogenic pressure
- 2 **POLICY SUCCESS:** Namami Gange has achieved 42–45% BOD reduction at 3 stations — STP commissioning is measurably working
- 3 **POLICY FAILURE:** Fecal Coliform has INCREASED 623–8055% — STPs lack tertiary disinfection. The bacterial crisis is worsening.
- 4 **ASSIMILATIVE PARADOX:** 18× discharge expansion dilutes BOD concentration but fails to assimilate pathogen loading — "the solution to pollution is NOT dilution"
- 5 **MONITORING GAP:** Annual min/max monitoring cannot detect acute festival shocks — India needs continuous real-time WQ sensors
- 6 **ONLY 1/5 SAFE:** Haridwar is the only station where bathing is safe by regulatory standards. 80% of major stations fail.

Future Work & Policy Recommendations

Future Research Directions

- Seasonal analysis: Pre-monsoon vs post-monsoon water quality (requires daily/monthly data from continuous sensors)
- OSM riparian mapping: Count cremation grounds, nala mouths, and industrial units within 2 km buffer per station
- Antibiotic resistance gene (ARG) profiling in river sediments
- Machine learning forecasting of WQI trends under climate scenarios

Policy Recommendations

- MANDATORY tertiary disinfection (UV/ozone) at all Namami Gange STPs
- Deploy continuous real-time water quality sensors at pilgrimage sites
- Enforce environmental flow (e-flow) releases during lean season
- Implement Faecal Sludge & Septage Management (FSSM) in riverside towns
- Bio-toilet mandates at all major ghats and during festival gatherings

References & Acknowledgements

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Acknowledgements: We thank CPCB for making national water quality data publicly available, the WorldPop Project for open demographic rasters, and the WWF/USGS for the HydroSHEDS open hydrographic dataset.

Thank You! Questions

Full repository, data, and reproducible code: github.com/prasad-raminen/Ganga-River-stem_ES-Project